

ZEYA, Russian Federation

WMO: 313000

Lat: 53.7421N

Lon: 127.2760E

Elev: 230

StdP: 98.59

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.7	(c) -33.6	(d) -37.8	(e) 0.1	(f) -35.3	(g) -35.8	(h) 0.1	(i) -33.0	(j) 6.7	(k) -11.4	(l) 5.9	(m) -13.1	(n) 0.0	(o) 220	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.4	(c) 30.2	(d) 20.2	(e) 28.5	(f) 19.3	(g) 26.8	(h) 18.8	(i) 21.9	(j) 27.4	(k) 21.0	(l) 26.2	(m) 20.0	(n) 24.9	(o) 1.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 20.2	(b) 15.3	(c) 24.4	(d) 19.2	(e) 14.4	(f) 23.5	(g) 18.2	(h) 13.5	(i) 22.6	(j) 65.6	(k) 27.4	(l) 62.1	(m) 26.1	(n) 58.7	(o) 24.9	(p) 25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.8	(b) 4.9	(c) 4.2	(e) -36.8	(f) 33.3	(g) 2.3	(h) 1.9	(i) -38.4	(j) 34.7	(k) -39.7	(l) 35.8	(m) -41.0	(n) 36.9	(o) -42.6	(p) 38.3		
			DB													
			WB													
				-36.8	23.5	2.2	0.9	-38.5	24.1	-39.8	24.7	-41.0	25.2	-42.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.6	-23.5	-18.5	-8.6	2.6	10.9	17.9	20.7	17.8	10.3	0.0	-14.5	-23.1
	DBStd	16.56	5.53	5.07	5.81	4.33	3.93	3.52	2.82	3.15	4.21	5.37	6.57	5.78
	HDD10.0	4801	1038	798	578	227	36	1	0	1	47	312	736	1027
	HDD18.3	7058	1297	1032	837	473	232	48	11	47	242	569	986	1285
	CDD10.0	938	0	0	0	3	64	238	332	242	57	2	0	0
	CDD18.3	153	0	0	0	0	2	36	84	31	1	0	0	0
	CDH23.3	1468	0	0	0	2	54	440	702	261	9	0	0	0
	CDH26.7	395	0	0	0	0	9	125	214	47	0	0	0	0
Wind	WSAvg	1.3	0.9	1.2	1.7	2.1	1.9	1.4	1.2	1.1	1.2	1.4	1.1	1.0
Precipitation	PrecAvg	527	4	4	8	27	35	85	142	106	73	24	12	7
	PrecMax	745	15	17	29	58	90	205	327	217	191	143	25	32
	PrecMin	361	0	0	0	5	5	12	39	26	14	1	3	0
	PrecStd	102	4	4	7	15	19	48	70	52	43	27	7	6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.4	-3.3	8.1	21.2	28.0	32.3	33.7	30.1	24.7	16.7	1.2	-3.8
		MCWB	-8.3	-5.9	2.3	10.2	15.5	19.7	20.9	21.1	17.1	9.9	-1.2	-6.3
	2%	DB	-10.0	-5.7	5.0	16.5	24.6	29.9	30.8	27.9	21.9	13.1	-0.7	-6.8
		MCWB	-11.6	-7.9	0.0	7.7	13.4	18.9	20.9	19.9	15.3	7.3	-2.5	-8.5
	5%	DB	-12.5	-8.2	2.5	13.0	21.9	27.9	29.0	26.2	19.8	10.6	-2.4	-11.8
		MCWB	-13.7	-9.9	-1.6	5.8	12.0	17.9	20.4	19.2	13.9	5.6	-4.2	-13.0
	10%	DB	-14.8	-10.4	0.4	10.5	19.3	25.9	27.2	24.5	17.8	8.0	-4.8	-14.6
		MCWB	-15.7	-11.9	-2.9	4.1	10.6	17.2	19.8	18.4	12.8	3.7	-6.4	-15.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.3	-5.5	2.7	10.9	16.2	21.9	23.7	22.5	18.2	10.3	-0.4	-6.0
		MCDB	-6.6	-3.3	7.8	20.4	26.1	28.8	29.5	26.9	23.5	15.9	0.6	-3.8
	2%	WB	-11.5	-7.8	0.2	8.0	14.4	20.4	22.4	21.3	16.3	7.8	-2.3	-8.5
		MCDB	-10.0	-5.8	4.4	15.5	22.5	27.5	27.9	26.0	20.6	12.7	-0.9	-6.7
	5%	WB	-13.7	-9.9	-1.3	6.0	12.9	19.1	21.5	20.2	15.1	5.9	-4.3	-12.9
		MCDB	-12.5	-8.3	2.2	12.4	19.4	25.8	26.9	24.8	18.1	10.2	-2.4	-11.9
	10%	WB	-15.7	-11.9	-2.8	4.4	11.5	18.0	20.6	19.2	13.7	4.1	-6.3	-15.4
		MCDB	-14.8	-10.5	0.3	9.9	17.8	24.4	25.5	23.3	17.1	7.6	-4.8	-14.6
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.6	11.6	11.2	12.1	12.2	10.4	10.9	11.4	9.7	8.8	7.8
		MCDBR	9.5	13.1	12.1	16.4	17.7	16.1	14.7	14.2	14.3	13.0	9.1	7.9
		MCWBR	8.4	11.2	8.2	9.4	8.4	6.7	6.1	7.0	7.9	7.9	7.5	6.9
	5% WB	MCDBR	9.5	13.0	11.4	15.1	15.0	13.7	11.4	11.7	11.7	11.7	8.6	8.1
		MCWBR	8.4	11.2	7.9	8.9	7.5	6.4	5.6	6.3	6.8	7.5	7.3	7.1
Clear-Sky Solar Irradiance	taub	0.249	0.282	0.337	0.492	0.535	0.496	0.477	0.418	0.381	0.355	0.286	0.248	
	taud	2.155	2.112	2.014	1.779	1.810	2.021	2.127	2.287	2.305	2.191	2.133	2.098	
	Ebn at Noon	717	804	824	725	721	762	770	800	782	700	641	623	
	Edh at Noon	64	95	132	194	198	162	144	116	100	89	66	55	
All-Sky Solar Radiation	RadAvg	1.24	2.42	4.21	5.28	5.25	6.07	5.22	4.58	3.46	2.23	1.35	0.92	
	RadStd	0.05	0.07	0.14	0.47	0.35	0.52	0.56	0.35	0.27	0.16	0.08	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page